Otto-Von-Guericke Universität, Magdeburg

Bosch Manufacturing Solutions GmbH

Summy Report on

Praktikum im Simulation Materialfluss von Montage- und Prüfanlagen

Submitted to fulfilment of

Compulsory internship as a part of M.Sc. Systems Engineering for Manufacturing

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Acknowledgement

This internship report documents a transformative journey within the realm of Material Flow Simulation, undertaken at BOSCH Manufacturing Solutions GmbH. Under the mentorship of Sebastian Schuetz and Felix Guehr, this experience has been an exploration of critical concepts and practical applications in industrial engineering. The report delves into the comprehensive training provided through the Example Project, addressing bottlenecks, and optimizing material flow in Customer Project A, and the complexities of a dual-product production line in Customer Project B. The acquired knowledge and expertise have cemented the importance of simulation in material flow optimization. The report concludes with a strong commitment to contribute to efficiency and innovation in the manufacturing industry.

Declaration

I, Akshay Chandrola, hereby declare that this internship report has been prepared by me as part of my internship at BOSCH Manufacturing Solutions GmbH. The report is an accurate and truthful representation of the work I performed during the internship period.

I further declare that any information, data, or content presented in this report has been obtained from reliable sources and has been used in accordance with academic and ethical standards.

To maintain confidentiality and protect proprietary information, it is essential to note that the names of customers and specific product details have been intentionally omitted within this report. This has been done to uphold the confidentiality agreements and commitments between BOSCH Manufacturing Solutions GmbH and its respective clients.

I acknowledge that the use of such confidentiality measures is a fundamental aspect of ethical business conduct, and I am committed to respecting and upholding these agreements.

By submitting this report, I affirm that all the work contained herein is original, and any external sources used have been appropriately cited and referenced. I understand the importance of adhering to the highest standards of integrity and ethics in my academic and professional endeavors.

Akshay Chandrola

Stuttgart

03/11/2023

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Introduction

I am delighted to present this internship report, a culmination of my enriching experience as an intern at BOSCH Manufacturing Solutions GmbH. This report represents an important milestone in my academic journey and professional development and is being submitted to both my esteemed university and my dedicated supervisors at BOSCH.

During my internship, I had the privilege of working in the field of Material Flow Simulation, a realm of strategic significance for optimizing manufacturing and production processes. This internship experience was made possible through the guidance, support, and mentorship of two exceptional individuals, Mr. Sebastian Schuetz and Mr. Felix Guehr. Their unwavering commitment to my growth and their expertise in the field of Material Flow Simulation were instrumental in shaping this journey.

Material Flow Simulation is a dynamic and evolving discipline that plays a pivotal role in enhancing operational efficiency, resource management, and overall productivity within the manufacturing industry. My internship at BOSCH not only provided me with a unique opportunity to gain hands-on experience but also allowed me to delve into the intricacies of this field. As the manufacturing landscape continues to evolve with increasing complexity and global competition, the significance of material flow simulation in streamlining processes and minimizing costs becomes ever more pronounced.

This report aims to encapsulate the knowledge and insights I acquired during my time at BOSCH, as well as the practical contributions I made to the field. Through this document, I hope to convey the invaluable experience I gained, the challenges I encountered, and the achievements I accomplished during my internship.

In the pages that follow, I will provide a detailed account of my roles and responsibilities, the methodologies I employed, the data I collected and analyzed, and the findings and recommendations that emerged from my work. It is my aspiration that this report serves as a testament to the commitment and dedication with which I approached my internship, as well as a source of knowledge and inspiration for future endeavors in Material Flow Simulation.

I extend my heartfelt gratitude to BOSCH Manufacturing Solutions GmbH, my university, and especially to Mr. Sebastian Schuetz and Mr. Felix Guehr for their unwavering support and mentorship. I trust that this report will reflect the depth of my learning and the value I gained from this remarkable internship experience.

Company Overview: BOSCH Manufacturing Solutions GmbH

BOSCH Manufacturing Solutions GmbH, a subsidiary of the globally renowned Bosch Group, stands as a leading global supplier in the realm of special purpose machinery. The company's mission revolves around offering tailor-made, intelligent assembly and testing solutions, advanced manufacturing processes, cutting-edge technologies, and an array of digital and intelligent services that span the entire product life cycle [1].

Expertise and Global Presence: BOSCH Manufacturing Solutions leverages its proficiency in international project management and the remarkable competence of its nine locations distributed across the globe. This unique combination allows the company to "think globally and act locally," providing the capability to harmonize the benefits of global standardization with the nuances of regional, market-specific conditions. The company's worldwide reach, and localized approach empower it to effectively address the diverse needs of its clients while ensuring top-tier solutions that align with the specific requirements of each market [1].

Commitment to Innovation and Sustainability: To meet the ever-evolving and future-oriented demands of its customers, BOSCH Manufacturing Solutions GmbH maintains an unwavering commitment to the continuous development of manufacturing processes and technologies. This dedication to innovation enables the company to stay at the forefront of the industry, delivering solutions that are both technologically advanced and environmentally sustainable [1].

In a dynamic and highly competitive manufacturing landscape, BOSCH Manufacturing Solutions upholds a reputation for excellence, quality, and customer-centricity. Its comprehensive suite of services is designed to optimize production efficiency, reduce costs, and enhance the overall product development life cycle [1].

Through this report, the reader is invited to explore the practical application of the company's principles, the significance of its contributions, and the collaborative spirit that

defines BOSCH Manufacturing Solutions GmbH as a leader in special purpose machinery and advanced manufacturing solutions [1].

Objectives and Scope of the Internship

The primary objectives of the internship conducted at BOSCH Manufacturing Solutions GmbH encompassed a comprehensive exploration of material flow simulation and its applications in the context of the company's specialized manufacturing solutions. The following objectives were outlined to guide the intern in achieving the desired outcomes:

1. Gain Practical Experience: To gain practical exposure to the field of material flow simulation, with a specific focus on its relevance and applications within the special purpose machinery manufacturing industry.

2. Apply Theoretical Knowledge: To apply theoretical concepts and knowledge acquired during academic coursework in a real-world industrial setting, effectively bridging the gap between classroom learning and practical application.

3. Contribute to Project Implementation: To actively contribute to ongoing material flow simulation projects under the guidance of experienced supervisors, Sebastian Schuetz and Felix Guehr, by actively participating in project development, data collection, analysis, and simulation exercises, including the creation of simulation models based on design.

4. Develop Analytical Skills: To enhance analytical and problem-solving skills by participating in data-driven decision-making processes and utilizing simulation software and tools to optimize manufacturing processes.

5. Foster Collaborative Skills: To develop effective teamwork and communication skills by actively engaging with cross-functional teams and project stakeholders to achieve common goals in material flow simulation projects.

6. Gain Industry Insights: To gain insights into the operational intricacies of BOSCH Manufacturing Solutions and the manufacturing industry, fostering a deeper understanding of the company's unique challenges and opportunities.

7. Demonstrate Adaptability: To adapt to the dynamic and evolving manufacturing landscape by embracing emerging technologies and concepts in material flow simulation, thereby contributing to the company's future-oriented approach.

8. Data Collection: The intern actively engaged in the collection of data related to material flow, production processes, and manufacturing systems within BOSCH Manufacturing Solutions.

9. Analysis and Simulation: Utilizing software and tools specific to material flow simulation, the intern performed data analysis and developed simulation models to optimize manufacturing processes. This included the creation of simulation models based on design specifications and project requirements.

10. Collaboration: The intern collaborated closely with supervisors and cross-functional teams, actively participating in project meetings and communication with project stakeholders.

11. Reporting: Regular reporting of project progress, findings, and outcomes to supervisors and the organization.

This internship aimed to offer a comprehensive hands-on experience in the field of material flow simulation, aligning with the objectives while ensuring the attainment of meaningful outcomes for both the intern and BOSCH Manufacturing Solutions GmbH. The scope encompassed all activities and responsibilities associated with the internship project under the guidance of the internship supervisors, including the creation of simulation models based on design.

Important concepts and definitions related to material flow simulation

This section provides essential insights into fundamental concepts and definitions in material flow simulation, serving as a cornerstone for the subsequent exploration of practical applications.

- **Material Flow Simulation**: Material flow simulation is a computer-based modeling and analysis technique used to replicate and analyze the movement of materials

within a manufacturing or logistical system. It enables organizations to optimize resource utilization, reduce bottlenecks, and enhance overall efficiency by creating digital representations of physical systems [2].

- **Bottleneck**: A bottleneck is a point in the production or material flow process where the rate of output is limited by the slowest or least efficient workstation, machine, or process. Identifying and mitigating bottlenecks is a key objective of material flow simulation [3].
- **Throughput**: Throughput refers to the rate at which products or materials flow through a production system. It is a crucial performance metric in material flow simulation, and improving throughput is often a primary goal [3].
- **Cycle Time**: Cycle time is the time required for a product to move through the entire production process, from the initial input stage to the final output. Reducing cycle time can lead to increased productivity and cost savings [3].
- **Process Mapping**: Process mapping involves creating visual representations of a production process or material flow, often using flowcharts or diagrams. It helps in understanding the sequence of activities and identifying areas for improvement [3].
- **Resource Allocation**: Resource allocation involves assigning machines, labor, and other resources to specific tasks and processes within the production system. Effective resource allocation is essential for optimizing material flow [3].
- **Just-in-Time (JIT)**: Just-in-Time is a manufacturing philosophy and strategy aimed at producing and delivering products precisely when they are needed, minimizing inventory and waste. Material flow simulation often supports JIT principles [4].
- **Lean Manufacturing**: Lean manufacturing is a production approach that focuses on minimizing waste, reducing lead times, and improving overall efficiency. Material flow simulation can be used to implement lean principles in manufacturing [5].
- **Queueing Theory**: Queueing theory is a mathematical concept used in material flow simulation to study and analyze waiting lines, such as those in production systems. It helps in understanding and optimizing waiting and processing times [6].
- **Constraint Management**: Constraint management involves identifying and managing constraints or bottlenecks in a production system to ensure the smooth flow of materials and products.

- **Simulation Software**: Simulation software is a critical tool used in material flow simulation. It includes programs and platforms that create digital models of production systems to evaluate performance and test potential improvements.
- **Kaizen**: Kaizen is a Japanese term that means "continuous improvement." It represents the practice of continuously seeking small, incremental improvements in processes and operations, which is often a core principle in material flow optimization [4].

These concepts and definitions are fundamental to understanding and implementing material flow simulation to improve manufacturing and logistics processes. They provide the foundation for analyzing and optimizing material flow within various industries.

Key concepts and objects commonly used in Siemens Plant Simulation

In Siemens Plant Simulation, a simulation and modeling software used for industrial planning and optimization, various concepts and objects are used to model and simulate real-world manufacturing and logistics systems. Below are explanations of some of the key concepts and objects commonly used in Siemens Plant Simulation:

1. Supermarket: In Siemens Plant Simulation, a "supermarket" represents a storage or buffer area within a production system. It's often used to store parts or products temporarily, allowing for a smoother flow of materials and reducing the impact of variability in production rates. Supermarkets are strategically placed within the production line to prevent bottlenecks and ensure a steady material flow. The image below displays the supermarket concept [3].

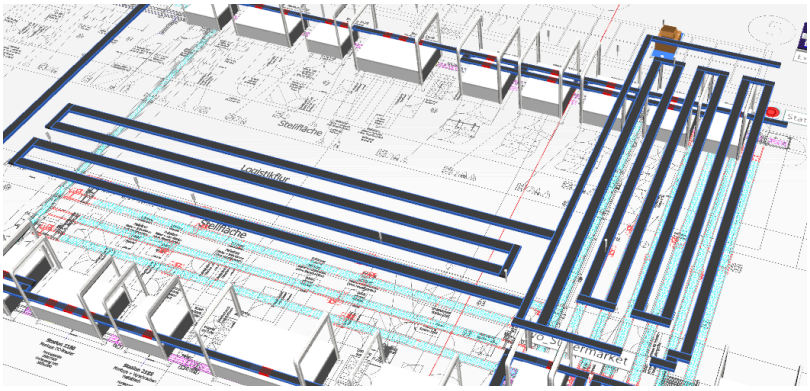


Figure 1: Supermarket Concept [7]

2. Lift: A "lift" in Siemens Plant Simulation refers to an automated material handling device used to move items between different elevations within a facility. Lifts are commonly used in multi-story warehouses or manufacturing plants to transport products or components efficiently. The image below depicts the lift concept in action.

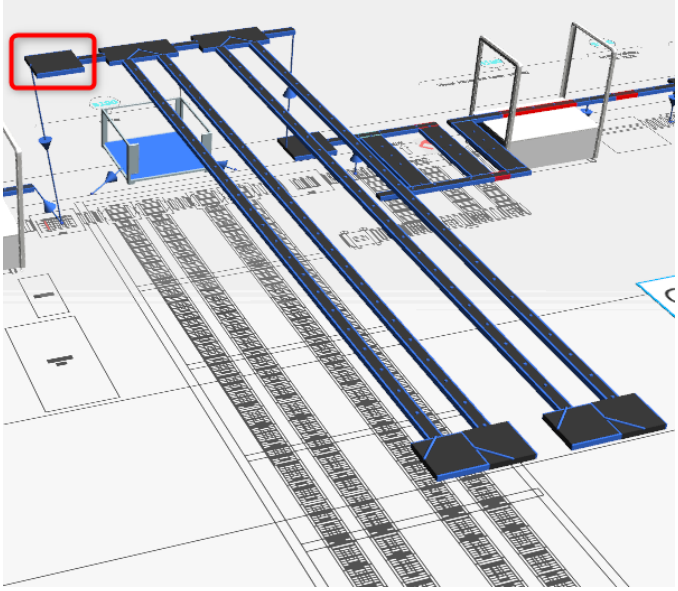


Figure 2: Lift Concept [7]

3. Buttons: In the context of Siemens Plant Simulation, "buttons" typically refer to user interface elements that allow users to control and interact with the simulation model. These buttons can trigger various actions or scenarios, such as starting simulations, changing parameters, or initiating specific processes. They provide a way for users to actively engage with and manipulate the simulation [3].

4. Display: The "display" object is used to present information and visualization within the simulation model. It can be used to show real-time data, key performance indicators (KPIs), or graphical representations of the system's state. Displays are often used to provide users with a visual understanding of the simulation's progress and performance [3].

5. Conveyor Systems: Conveyor systems are used to transport materials, products, or components within a simulation model. They represent the physical movement of items through the production or logistics system. Siemens Plant Simulation offers various types of

conveyor systems, including belt conveyors, roller conveyors, and automated guided vehicles (AGVs) [3].

6. Resource: A "resource" in Siemens Plant Simulation represents a work center, machine, or human operator involved in the production process. Resources are essential for modeling and simulating the labor and equipment involved in manufacturing or logistics operations [3].

7. Process Flow Diagram: A "process flow diagram" is a visual representation of the flow of materials or products through a system. It illustrates the sequence of operations, workstations, and interactions within the simulated environment. Process flow diagrams help users understand and optimize the production or logistic process [3].

8. Events and Triggers: Events and triggers in Siemens Plant Simulation allow for the introduction of dynamic elements into the simulation model. Events can be used to simulate disruptions, delays, or other unexpected occurrences that might impact the system's performance. Triggers are conditions that initiate specific actions or responses within the simulation [3].

9. Workstation: A workstation is a specific location or area within the production or material flow system where tasks are performed on the materials, products, or components. Workstations are equipped with tools and equipment required for these tasks.

10. Material Handling Equipment: Material handling equipment includes various devices, such as forklifts, pallet jacks, and automated robotic arms, used for transporting, lifting, and positioning materials and products in a material flow system [3].

11. Automated Guided Vehicle (AGV): An AGV is a mobile robot that autonomously follows predefined paths or routes to transport materials and components within a factory or warehouse. AGVs are commonly employed to automate material handling processes [3].

12. Assembly Line: An assembly line is a production system where individual components or subassemblies are progressively added to a product as it moves along a predetermined path. It is a common approach in manufacturing to streamline the assembly process [3].

13. Manual Station: In Plant Simulation, a "manual station" represents a workstation where human operators perform manual tasks in the simulated production or logistics system. The figure below depicts a snapshot of the manual station in the simulation.

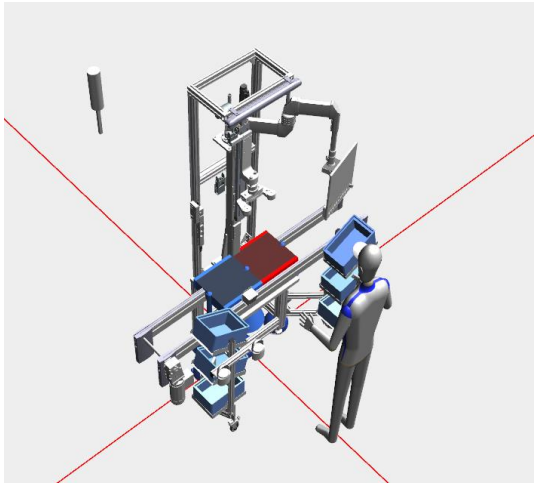
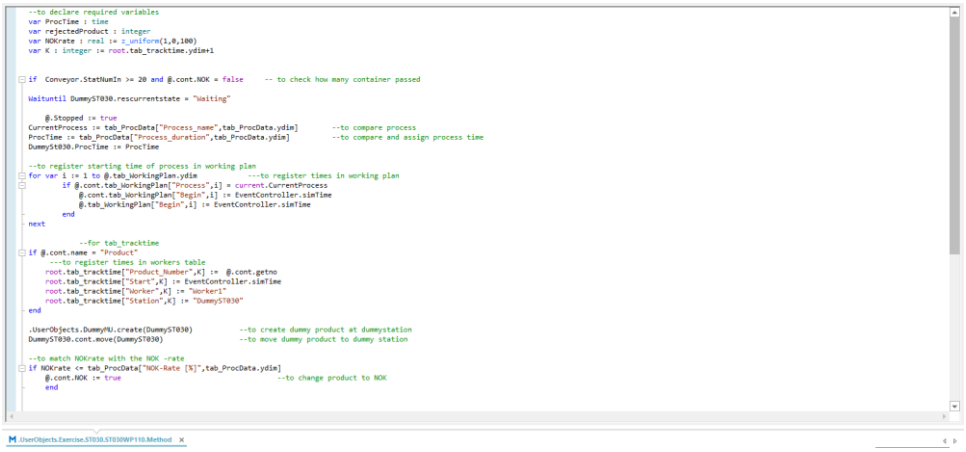


Figure 3: Manual Station [7]

14. Methods: Methods in Plant Simulation are scripts that customize and automate object behaviors, facilitating dynamic event handling and system optimization by defining how objects respond to changing conditions and triggers. The figure below depicts a visual representation of the Method Concept [3].



```
--to declare required variables
var ProcTime : time
var rejectedProduct : integer
var NOKrate : real := z_uniform(1,0,100)
var K : integer := root.tab_tracktime.ydim+1


if  Conveyor.StatNumIn >= 20 and @.cont.NOK = false     -- to check how many container passed

Waituntil DummyST030.rescurrentstate = "Waiting"

    @.Stopped := true
CurrentProcess := tab_ProcData["Process_name",tab_ProcData.ydim]        --to compare process
ProcTime := tab_ProcData["Process_duration",tab_ProcData.ydim]          --to compare and assign process time
DummyST030.ProcTime := ProcTime

--to register starting time of process in working plan
for var i := 1 to @.tab_WorkingPlan.ydim            ---to register times in working plan
        if @.cont.tab_WorkingPlan["Process",i] = current.CurrentProcess
            @.cont.tab_WorkingPlan["Begin",i] := EventController.simTime
            @.tab_WorkingPlan["Begin",i] := EventController.simTime
        end
next

            --for tab_tracktime
if @.cont.name = "Product"
     ---to register times in workers table
    root.tab_tracktime["Product_Number",K] :=  @.cont.getno
    root.tab_tracktime["Start",K] := EventController.simTime
    root.tab_tracktime["Worker",K] := "Worker1"
    root.tab_tracktime["Station",K] := "DummyST030"
end

.UserObjects.DummyMU.create(DummyST030)            --to create dummy product at dummystation
DummyST030.cont.move(DummyST030)                   --to move dummy product to dummy station

--to match NOKrate with the NOK -rate
if NOKrate <= tab_ProcData["NOK-Rate [%]",tab_ProcData.ydim]
    @.cont.NOK := true                                  --to change product to NOK
    end
```

Figure 4: Method Concept [7]

These definitions cover key components and concepts relevant to material flow simulation and material flow systems, helping to optimize processes and enhance efficiency in various industrial settings.

Project Experiences

This section provides detailed insights into three significant projects undertaken during the internship period. Each project offers a unique learning experience and practical exposure to material flow simulation. The Example Project serves as the foundation, while Customer Project 1 and Customer Project 2 are actual projects conducted for specific customers, with their names withheld for confidentiality purposes. The following subsections delve into the specifics of each project, their objectives, challenges, and outcomes.

Example Project: Material Flow Simulation Training

The Example Project serves as a comprehensive training program aimed at equipping participants with the essential skills and knowledge required to successfully conduct material flow simulation projects. This project covers a wide range of concepts and practices related to simulation, making it an invaluable foundation for real-world projects. The figure below illustrates one of the examples from the Example Project.

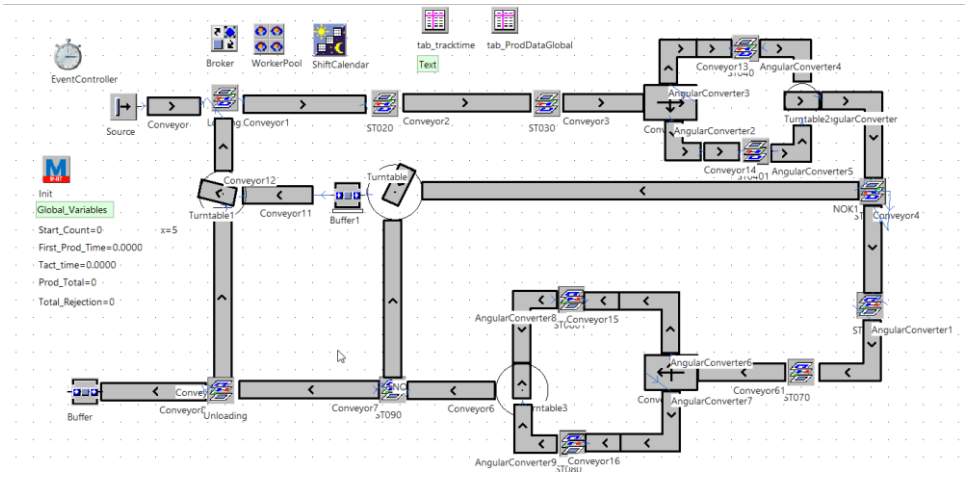


Figure 5: Example Project [7]

This training program provides participants with a solid foundation in material flow simulation, ensuring that they are well-prepared to contribute to actual projects. The step-by-step tasks offer a comprehensive learning experience.

Customer Project A: Production Line Optimization

Project Description: Customer Project A is a confidential material flow simulation project undertaken for a specific customer facing a bottleneck issue within their existing production line for Product A. The customer's goal is to improve the situation by introducing a supermarket to the production line and evaluating different scenarios to find the most efficient implementation. The project involves a user interface with various buttons, as shown in the attached figure, to control and test different scenarios. The figure below represents a snapshot of Customer Project A [7].

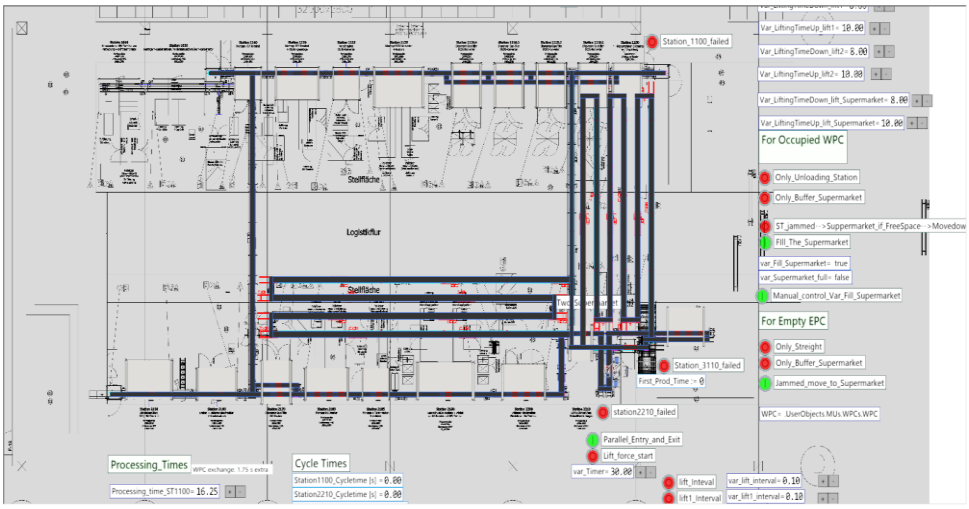


Figure 6: Customer Project A [7]

Production Line Bottleneck: The customer's production line for Product A is currently experiencing a bottleneck, which is a point in the production process where the rate of output is limited by the slowest or least efficient workstation or machine. This bottleneck results in inefficiencies, delays, and reduced overall productivity.

Objective: The primary objective of Customer Project A is to address the bottleneck issue in the production line for Product A by introducing a supermarket, which serves as a buffer for components or products. The project aims to find the best way to implement the supermarket into the existing line, allowing for smoother material flow, reduced wait times, and increased throughput [7].

Key Project Components: The project involves the following key components:

- **Supermarket Implementation**: The introduction of a supermarket as a buffer on the production line to temporarily store products or components.
- **Scenario Testing**: The use of a user interface with different buttons to control and test various scenarios for implementing the supermarket.
- **Material Flow Simulation**: Utilization of material flow simulation software to model the current production line, simulate different scenarios, and analyze the effects of the supermarket introduction.
- **Data Collection**: Gathering data on the current production line's performance, including cycle times, throughput, and resource utilization.

User Interface and Scenarios: The user interface provided for the project includes multiple buttons, each corresponding to a different scenario. These scenarios may involve varying the supermarket's location, size, or capacity. By clicking these buttons, the customer can initiate simulations to evaluate each scenario's impact on the production line's performance [7].

Simulation and Analysis: The project involves the use of material flow simulation software to create digital models of the existing production line and different scenarios with the added supermarket. Simulation allows for the observation of key performance indicators (KPIs) such as throughput, cycle times, and resource utilization. These KPIs will be analyzed to determine the most efficient implementation of the supermarket.

Expected Outcomes: Customer Project A aims to provide the customer with a data-driven recommendation for the best way to implement the supermarket in their production line for Product A. The expected outcomes include optimized material flow, reduced bottlenecks, increased efficiency, and improved production performance [7].

Customer Project B: Dual-Product Production Line Implementation

Project Description: Customer Project B is a confidential material flow simulation project designed to address the customer's need for a concept production line capable of producing two different products: Product A and Product B. Both products have distinct process requirements, and the project aims to create an efficient line that can accommodate these variations. The figure below represents a scenario from Customer Project B [7].

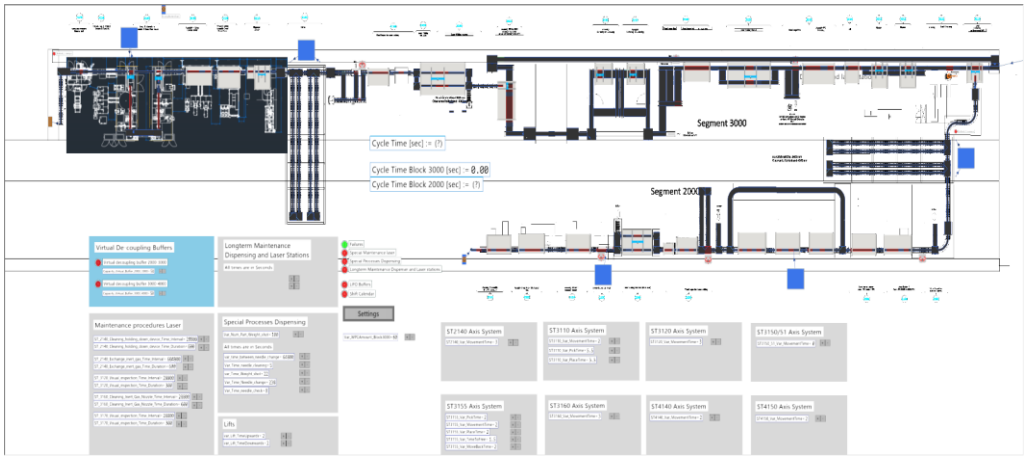


Figure 7: Customer Project B [7]

Dual-Product Production Line Challenge:

The challenge in this project is to design a production line that can seamlessly handle the manufacturing of two different products, each with its own set of process requirements. The goal is to optimize the line's flexibility, resource utilization, and overall efficiency to ensure that both Product A and Product B can be produced without disruptions [7].

Objective:

The primary objective of Customer Project B is to create a dual-product production line concept that accommodates the production of Product A and Product B, while ensuring that the unique process requirements for each product are met. This includes designing a flexible layout, workflow, and resource allocation strategy [7] .

Key Project Components:

The project encompasses several key components, including:

- **Production Line Layout:** The design and arrangement of workstations, machinery, and resources in the production line to accommodate both products efficiently.
- **Process Requirements:** Understanding and defining the specific process requirements for Product A and Product B, including any differences in workflow, assembly, or testing.
- **Resource Allocation:** Determining how resources (such as machines, labor, and materials) will be allocated and scheduled to meet the production needs of both products.

- **Simulation and Modeling:** Using material flow simulation software to model the dual-product production line, test different scenarios, and evaluate the line's performance.

Dual-Product Production Line Design:

The project will focus on creating a production line layout that allows for the seamless transition between producing Product A and Product B. This includes considerations for changeovers, equipment flexibility, and resource adaptability to accommodate both products' specific requirements [7].

Simulation and Analysis:

Material flow simulation software will be employed to build digital models of the dual-product production line concept. The simulation will involve running scenarios for producing both Product A and Product B, assessing cycle times, throughput, and resource utilization. These analyses will inform decision-making and optimization strategies.

Expected Outcomes:

Customer Project B aims to deliver a concept for a dual-product production line that is adaptable and efficient, meeting the unique process requirements of both Product A and Product B. The expected outcomes include a well-defined layout, resource allocation strategy, and a simulation-based understanding of how the line can successfully produce both products [7].

The project will provide the customer with valuable insights into the design and operation of a dual-product production line, optimizing resource utilization and production efficiency. Ultimately, the goal is to enhance the customer's capability to manufacture both Product A and Product B with minimal disruptions and maximum efficiency.

Conclusion

In summary, this internship at BOSCH Manufacturing Solutions GmbH has been an enriching experience, equipping me with essential skills and knowledge in Material Flow Simulation. The Example Project provided a strong foundation, while the practical exposure in Customer Projects A and B underscored the real-world significance of efficient material flow.

Guided by mentors Sebastian Schuetz and Felix Guehr, this internship has nurtured a commitment to continuous learning, adaptability, and the application of simulation to address industrial challenges. It bridges the gap between academia and industry, setting the stage for a promising future in material flow optimization and manufacturing efficiency.

I am grateful for the opportunities and knowledge gained during this internship, and I look forward to contributing to innovation and efficiency in the manufacturing industry in the years to come.

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